A Stability Calculus for Local Learning Rules

1 Definitions and Preliminaries

1.1 Data Model and Sample

Definition 1 (Sample)**.** A *sample* S of size n is an ordered tuple

$$S = \big((x_1, y_1), \ldots, (x_n, y_n)\big)$$

where each $x_i \in X$ lies in a finite metric space (X, d) and each label $y_i \in \{0, 1\}$.

The order of the tuple is semantically relevant only for deterministic tie-breaking. Two identical labeled points appearing at different indices are treated as distinct sample occurrences.

- **Duplicate points are allowed.**
- **Conflicting labels at the same point are allowed.**

Any theorem or experiment that excludes duplicates or conflicts must state this explicitly.

Definition 2 (Finite Metric Space)**.** A finite metric space is a finite set X together with a distance function

$$d : X \times X \to \mathbb{R}_{\geq 0}$$

satisfying for all $x, x', z \in X$:

1. $d(x, x) = 0$;
2. $d(x, x') > 0$ for $x \neq x'$;
3. $d(x, x') = d(x', x)$;
4. $d(x, z) \leq d(x, x') + d(x', z)$.

Definition 3 (Graph Shortest-Path Metric)**.** For a finite, simple, connected, unweighted, undirected graph $G = (V, E)$, the graph metric is

$$d_G(u, v) = \text{length of a shortest path from } u \text{ to } v.$$

Disconnected graphs are rejected in code paths that require a metric.

1.2 Prediction Rule

The query point x may be any element of X, including a point that appears in the training sample.

Note 1 (Deterministic Tie-Breaking). Neighbor ordering is determined lexicographically by:

1. smaller distance to the query point;
2. smaller sample index in the ordered tuple.

If class votes tie after the first k ordered neighbors are selected, the predicted label is resolved by fixed label order $0 \prec 1$. Ties are always broken in favor of label 0.

Definition 4 (k-NN Prediction)**.** Given $k \in \mathbb{N}$ with $1 \le k \le n$, the deterministic k-NN predictor $h_S^{(k)}(x)$ is obtained by:

1. ordering the sample occurrences by the deterministic neighbor rule above;
2. selecting the first k occurrences;
3. taking the majority vote of their labels;
4. resolving label-vote ties by $0 \prec 1$.

1.3 Loss Function

Definition 5 (Binary 0-1 Loss)**.** For a hypothesis h and a labeled point $(x, y) \in X \times \{0, 1\}$, the binary classification loss is

$$\ell(h, (x, y)) = \mathbf{1}\{h(x) \neq y\}.$$

1.4 Sample Perturbations

Let $S = \big((x_1, y_1), \ldots, (x_n, y_n)\big)$.

Definition 6 (Delete-One)**.** For index $i \in \{1, \ldots, n\}$, define

$$S^{-i} = \big((x_1, y_1), \ldots, (x_{i-1}, y_{i-1}), (x_{i+1}, y_{i+1}), \ldots, (x_n, y_n)\big).$$

Definition 7 (Add-One)**.** For any labeled point $z = (x, y) \in X \times \{0, 1\}$, define

$$S \oplus z$$

as the ordered tuple obtained by appending z at the end.

Definition 8 (Replace-One)**.** For index i and labeled point $z = (x', y') \in X \times \{0, 1\}$, define

$$S^{i \leftarrow z}$$

as the ordered tuple obtained by replacing the i-th occurrence of S with z.

In v0.1 worst-case stability definitions, the replacement point ranges over all labeled points in $X \times \{0, 1\}$, not over a data-generating distribution.

1.5 Stability Notions

Unless otherwise stated, all pointwise notions are evaluated with fixed k, fixed metric space, fixed tie-breaking rule, and fixed loss defined above.

Note 2. Pointwise stability statements concern a fixed training sample, a fixed perturbation, and a fixed evaluation pair (x, y). Expected stability refers to the expectation under an explicitly specified data-generating distribution. Uniform stability is the supremum of the corresponding indicator over all valid ordered training tuples, all allowed perturbation choices, and all evaluation pairs.

Definition 9 (Pointwise Delete-One Stability Indicator)**.** For index i and query-label pair (x, y), define

$$\Delta_{\mathrm{del}}(S, i, x, y) = \left| \ell(h_S^{(k)}, (x, y)) - \ell(h_{S^{-i}}^{(k')}, (x, y)) \right|,$$

where $k' = \min(k, n - 1)$ when deletion reduces sample size below k.

Definition 10 (Pointwise Replace-One Stability Indicator)**.** For index i, replacement z, and query-label pair (x, y), define

$$\Delta_{\mathrm{rep}}(S, i, z, x, y) = \left| \ell(h_S^{(k)}, (x, y)) - \ell(h_{S^{i \leftarrow z}}^{(k)}, (x, y)) \right|.$$

Definition 11 (Pointwise Add-One Stability Indicator)**.** For added labeled point z and query-label pair (x, y), define

$$\Delta_{\mathrm{add}}(S, z, x, y) = \left| \ell(h_S^{(k)}, (x, y)) - \ell(h_{S \oplus z}^{(k)}, (x, y)) \right|.$$

Definition 12 (LOO / CVloo Stability)**.** For each index i, define

$$\Delta_{\mathrm{loo}}(S, i) = \left| \ell(h_{S^{-i}}^{(k')}, (x_i, y_i)) - \ell(h_S^{(k)}, (x_i, y_i)) \right|.$$

LOO evaluates at the deleted sample occurrence itself. This definition is occurrence-based: if the same point appears twice, deleting one copy and evaluating at that occurrence is distinct from deleting the other.

For LOO, the uniform stability supremum is over ordered tuples S and indices i, with the evaluation pair fixed to the deleted occurrence.

1.6 Consistency Note

The consistency notion used in paper-level discussion is classical distributional consistency for k-NN classification. It is not a finite-sequence surrogate notion. Any finite-metric experiment provides evidence about worst-case stability behavior; it does not by itself constitute a consistency statement.

References